Technische Universität Dresden

FTM Fügetechnik und Montage

Faculty of Mechanical Science and Engineering

Chair of Joining Technology and Assembly

Diploma Thesis

Der Titel der Arbeit

Vorname Nachname

Born on: 1st February 1990 in Dresden
Matriculation number: 123456789

to achieve the academic degree

Diplomingenieur (Dipl.-Ing.)

First referee

Prof. Dr.-Ing. Hans Christian Schmale

Second referee

2nd Referee

Supervisor

Supervisor I

Submitted on: 5th September 2026

Statement of authorship

I hereby certify that I have authored this document entitled *Der Titel der Arbeit* independently and without undue assistance from third parties. No other than the resources and references indicated in this document have been used. I have marked both literal and accordingly adopted quotations as such. There were no additional persons involved in the intellectual preparation of the present document. I am aware that violations of this declaration may lead to subsequent withdrawal of the academic degree.

Dresden, 5th September 2026

Vorname Nachname

Abstract

This is an abstract.

Zusammenfassung

Das ist eine Zusammenfassung.

Contents

List of Figures

List of Tables

Abbreviations

EDX	Energy Dispersive X-ray Spectroscopy

0 About this Template

Everything this thesis needs is in three files: `main.tex`, the class `ftmthesis.cls` and the bibliography `references.bib`. Images go into `figs/`. Delete this chapter once you have read it.

0.1 Language

Write `lang=en` or `lang=de` in the first line, whichever language you write in:

```
\documentclass[lang=de]{ftmthesis}
```

Add further options after it if you need them, for example `BCOR=10mm` for a wider binding correction or `oneside` if the thesis is not printed double-sided.

Use `\iflang{<german>}{<english>}` wherever a piece of text has to differ between the two languages.

0.2 Cover Page

Fill in the commands above `\maketitle`: `\title`, `\author`, `\date`, `\dateofbirth`, `\placeofbirth`, `\matriculationnumber`, `\supervisor` and `\referee`. Separate two referees with `\and`.

State the type of work with `\thesis` and `\graduation`:

```
\thesis{diploma}
\graduation[Dipl.-Ing.]{Diplomingenieur}
```

Instead of `diploma` you can write `bachelor`, `master`, `student`, `project` or `seminar`; the full list is in the `tudscr` documentation.

Faculty, chair and the FTM logo are already set, leave them alone.

0.3 Front Matter

Leave the `declarations` environment as it is, the statement of authorship needs no input.

Write the abstract twice, German first, English second:

```
\makeabstract
{Das ist eine Zusammenfassung.}
{This is an abstract.}
```

Keep or delete `\tableofcontents`, `\listoffigures` and `\listoftables` as you need them.

Declare every abbreviation above `\makeacronyms` and write `\gls{edx}` in the text:

```
\newacronym{edx}{EDX}{Energy Dispersive X-ray Spectroscopy}
```

The first use spells it out, Energy Dispersive X-ray Spectroscopy (EDX), every use after that is short, EDX. An abbreviation you never use does not appear in the list.

0.4 Main Matter

Write your chapters after `\mainmatter` with `\chapter` and `\section`. Give anything you want to point at a `\label` and refer to it with `\Cref`, which supplies the name itself, as in Chapter 0.

0.5 Figures and Tables

Put your images into `figs/` and scale them relative to `\linewidth`, see Figure 1. Write the `\caption` first and the `\label` after it.

Figure 1: A figure with a caption below it.

Build tables with `\toprule`, `\midrule` and `\bottomrule`, without vertical rules, see Table 1. Put the caption above the table.

Write displayed equations in an `equation` environment and use `\bm` for bold symbols:

$$E^2 = (\boldsymbol{p}\,c)^2 + \left(m_0\,c^2\right)^2\,, \qquad \gamma = \frac{1}{\sqrt{1 - \boldsymbol{v}^2/c^2}} \tag{0.1}$$

Equation (0.1) relates the energy of a particle to its momentum and its rest mass.

Table 1: A table with its caption above it.

Title 1	Title 2	Title 3
Element 1.1	Element 1.2	Element 1.3
Element 2.1	Element 2.2	Element 2.3
Element 3.1	Element 3.2	Element 3.3

0.6 Bibliography

Paste the BibTeX entry offered by the publisher or the search engine into `references.bib` and cite it by its key: `\cite{mathiszik2024non}` gives [1]. An entry you never cite is not printed.

0.7 Back Matter and Appendix

Write the acknowledgment after `\makeacknowledgment`.

Write the appendices after `\appendix`, each one as a `\chapter`, and they are lettered A, B and so on.

1 Introduction

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

2 Conclusion

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Bibliography

[1] Christian Mathiszik et al. “Non-destructive characterization of resistance projection welded joints by ultrasonic and passive magnetic flux density testing”. In: *Welding in the World* 68.10 (2024), pp. 2671–2682.

Acknowledgment

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A Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.